

Barton Springs Salamanders

Jamhad Byers & Nick Gonzalez

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0.1 Barton Springs Salamanders

The Barton Springs Salamanders DO and Flow data-set from the City of Austin provides measurements of dissolved oxygen (DO) and spring flow rates at Barton Springs, an important habitat for the endangered Barton Springs Salamander (*Eurycea sosorum*). The data-set includes time-series records of DO concentrations (in mg/L) and spring flow (in cubic feet per second) collected from various sites within the Barton Springs complex. These environmental factors are critical indicators of water quality and habitat health for the salamander population. By analyzing trends in DO levels and flow rates, researchers and conservationists can assess the impact of environmental changes, groundwater conditions, and human activity on the ecosystem.

This dataset is interesting do to its relevance in the broader conversation about climate change and the methods used to keep track of the nuanced and widespread effects. For our purposes here, the data will be used to asses dissolved oxygen levels, salamander counts, flow severity, and sample areas of four sights in the Austin area. Each row represents a measurement on a particular day at a particular site.

0.1.1 Original Variables

`site_name` is, as the name suggest, the name of the site for that particular instance of data collection.

`sample_date` is also as it's name suggest, in that it list the date that sample was taken.

`parameter result` & `unit` are the measurement, numeric value of that measurement, and the units of that measurement respectively

0.1.2 Created Variables

DO is created from the Dissolved Oxygen value in `PARMETER`, and contains the numeric values of the measurement.

Flowcode is created from the `FLOW SEVERITY CODE` (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY) value in `PARMETER`, and is categorical representation by the Texas Commission on Environmental Quality (TCEQ) of the type of flow observed. The numbers correspond to their listed type.

`salamanders` is created from the `NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED` value in `'PARMETER'`, and is a count of how many salamanders were seen during that particular measurement.

Finally, `area` is created from the `AREA SAMPLED` value in `PARMETER`, and is a measure of the total square footage that was survived for a particular site on a particular day.

The data is provided by the Barton Springs Salamanders DO and Flow data-set from the City of Austin and is a “view” of the Water Quality Sampling Data dataset - in that the Parameters and sites and projects are limited to those related to this particular subject (Barton Springs Salamanders, Flow and DO).

Before cleaning, the set contains 50,068 rows and 22 columns.

0.2 Methods

Our primary research questions are as follows;

- 1.) Does water Filter Fraction (`FILTER`), which refers to whether the water sample was filtered before it was analyzed, influence the Dissolved Oxygen (in mg/L) measurements?
- 2.) How do dissolved oxygen (DO) levels vary across different spring sites, and what is their relationship with seasonal changes?

0.2.1 Loading Packages

```
# Upload / install packages
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.1
library(plotROC)
```

0.2.2 Importing the Data

```
water <- read_csv("Barton_Springs_Salamanders_DO_and_Flow_20250305.csv")

## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)

# data preview
head(water)

## # A tibble: 6 x 22
##   WATERSHED  SAMPLE_DATE  SITE_NAME LON_DD_WGS84 LAT_DD_WGS84 SITE_TYPE MEDIUM
```

```
##   <chr>           <chr>           <chr>           <dbl>           <dbl> <chr>           <chr>
## 1 Barton Creek 02/25/2025 ~ Barton S~      -97.8           30.3 Spring Groun~
## 2 Barton Creek 01/08/2025 ~ Barton S~      -97.8           30.3 Spring Groun~
## 3 Barton Creek 12/11/2024 ~ Old Mill~      -97.8           30.3 Spring Groun~
## 4 Barton Creek 12/11/2024 ~ Barton S~      -97.8           30.3 Spring Groun~
## 5 Barton Creek 12/11/2024 ~ Eliza Sp~      -97.8           30.3 Spring Groun~
## 6 Barton Creek 11/21/2024 ~ Barton S~      -97.8           30.3 Spring Habit~
## # i 15 more variables: PARAM_TYPE <chr>, PARAMETER <chr>, QUALIFIER <lgl>,
## #   RESULT <dbl>, UNIT <chr>, FILTER <chr>, SAMPLE_ID <chr>,
## #   SAMPLE_SITE_NO <dbl>, METHOD <chr>, QC_FLAG <chr>, PROJECT <chr>,
## #   DATA_REF_NO <dbl>, SAMPLE_REF_NO <dbl>, TIME_NULL <chr>, QC_TYPE <lgl>
```

```
summary(water)
```

```
##   WATERSHED      SAMPLE_DATE      SITE_NAME      LON_DD_WGS84
## Length:50068    Length:50068    Length:50068    Min.   :-97.77
## Class :character Class :character Class :character 1st Qu.: -97.77
##   LAT_DD_WGS84  SITE_TYPE      MEDIUM      PARAM_TYPE
## Min.   :30.26    Length:50068    Length:50068    Length:50068
## 1st Qu.:30.26    Class :character Class :character Class :character
##   PARAMETER      QUALIFIER      RESULT      UNIT
## Length:50068     Mode:logical  Min.   : -5.50  Length:50068
## Class :character NA's:50068    1st Qu.:  0.00  Class :character
##   FILTER      SAMPLE_ID      SAMPLE_SITE_NO  METHOD
## Length:50068    Length:50068    Min.   : 35.0   Length:50068
## Class :character Class :character 1st Qu.: 183.0   Class :character
##   QC_FLAG      PROJECT      DATA_REF_NO    SAMPLE_REF_NO
## Length:50068    Length:50068    Min.   : 91172  Min.   : 9986
## Class :character Class :character 1st Qu.:1692903 1st Qu.:305031
##   TIME_NULL      QC_TYPE
## Length:50068     Mode:logical
## Class :character NA's:50068
## [ reached getOption("max.print") -- omitted 5 rows ]
```

0.2.3 Choosing 4 Sites for Analysis

```
# site options
unique(water$SITE_NAME)
```

```
## [1] "Barton Spring"
## [2] "Old Mill (Sunken Gardens) Spring"
## [3] "Eliza Spring"
## [4] "Old Mill Stream (outflow from Sunken Gardens Pool)"
## [5] "Upper Barton Spring"
## [6] "High Barton Spring"
```

Our analysis will focus on the Barton Spring, Eliza Spring, Old Mill (Sunken Gardens) Spring, and Upper Barton Spring sites.

0.2.4 Understanding Filter Fraction

```
unique(water$FILTER)
```

```
## [1] "Dissolved" "Total"      "N/A"
```

“Total” samples includes everything, from dissolved stuff tp particles floating in the sample

(suspended solids). “Dissolved” samples only include the dissolved substances, the sample is passed through a filter first, which removes solids. “N/A” is currently unknown.

0.2.5 Choosing Parameters to Review

The paramters variable is a representation of what measurement was taken for that row. Here, we display all available measurements for analysis in the dataset

```
# list of parameters
unique(water$PARAMETER)

## [1] "DISSOLVED OXYGEN"
## [2] "TOTAL TIME SPENT"
## [3] "NUMBER OF BARTON SPRINGS SALAMANDERS PHOTOGRAPHED"
## [4] "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED"
## [5] "FLOW"
## [6] "AUSTIN BLIND SLMNDR (1-2IN.)"
## [7] "AUSTIN BLIND SLMNDR (>=2IN.)"
## [8] "BARTON SPRINGS SLMNDR (TOTAL 1-2IN.)"
## [9] "BARTON SPRINGS SLMNDR (TOTAL >=2IN.)"
## [10] "BARTON SPRINGS SLMNDR (TOTAL <1IN.)"
## [11] "AUSTIN BLIND SLMNDR (<1IN.) (JUVENILE)"
## [12] "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)"
## [13] "SUM OF BSS SALAMANDERS PER TRANSECT"
## [14] "SUM OF ABS SALAMANDERS PER TRANSECT"
## [15] "BASEFLOW"
## [16] "AUSTIN BLIND SALMNR (ADULT) > 1 INCH"
## [17] "BARTON SPGS SALMNR (ADULT) > 1 INCH"
## [18] "SALAMANDER (UNKNOWN LENGTH)"
## [19] "SALAMANDERS PHOTGRAPHED >2 INCHES"
## [20] "SALAMANDERS PHOTGRAPHED <1 INCH"
## [21] "SALAMANDERS PHOTGRAPHED 1-2 INCHES"
## [22] "SALAMANDERS NOT PHOTGRAPHED >2 INCHES"
## [23] "SALAMANDERS NOT PHOTGRAPHED <1 INCH"
## [24] "SALAMANDERS NOT PHOTGRAPHED 1-2 INCHES"
## [25] "AUSTIN BLIND SALAMANDER (UNKNOWN LENGTH)"
## [26] "BARTON SPRINGS SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [27] "BARTON SPRINGS SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [28] "BARTON SPRINGS SLMNDR (1-2IN.) IN VEGETATION/MOSS"
## [29] "BARTON SPRINGS SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [30] "AUSTIN BLIND SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [31] "AUSTIN BLIND SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [32] "AUSTIN BLIND SLMNDR (1-2IN.) IN LEAF LITTER"
## [33] "AUSTIN BLIND SLMNDR (>=2IN.) IN LEAF LITTER"
## [34] "AUSTIN BLIND SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [35] "AUSTIN BLIND SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [36] "AUSTIN BLIND SLMNDR (1- 2IN.) IN VEGETATION/MOSS"
## [37] "AUSTIN BLIND SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [38] "BARTON SPRINGS SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [39] "BARTON SPRINGS SLMNDR (<1IN.) IN LEAF LITTER"
## [40] "BARTON SPRINGS SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [41] "AUSTIN BLIND SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [42] "AUSTIN BLIND SLMNDR (<1IN.) IN LEAF LITTER"
## [43] "AUSTIN BLIND SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [44] "AUSTIN BLIND SLMNDR (<1IN.) IN VEGETATION/MOSS"
```

```
## [45] "BARTON SPRINGS SLMNDR (<1IN.) IN VEGETATION/MOSS"
## [46] "BARTON SPRINGS SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [47] "BARTON SPRINGS SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [48] "BARTON SPRINGS SLMNDR (1-2IN.) IN LEAF LITTER"
## [49] "BARTON SPRINGS SLMNDR (>=2IN.) IN LEAF LITTER"
## [50] "AREA SAMPLED"
## [ reached getOption("max.print") -- omitted 12 entries ]
```

The definitions of these were provided by Mr. Robert Clayton from the City of Austin Watershed Protection Department. For this analysis, the measurements of interest include DISSOLVED OXYGEN, FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY), NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED, and AREA SAMPLED.

0.2.6 Associated Units List

The units of interest all correspond to the measurements of interest, so we removed all unnecessary unit types from the dataset for analysis

```
unique(water$UNIT)
```

```
## [1] "MG/L"
## [2] "Minutes"
## [3] "Count"
## [4] "Cubic Feet Per Second"
## [5] "None"
## [6] "1-2: (Yes; No)"
## [7] "Feet per second"
## [8] "(%) Percent"
## [9] "Sq Ft"
## [10] "Inches"
## [11] "Feet"
## [12] "MM"
## [13] "gal/min"
## [14] "Relative Abundance (1): <20 (2): 21-50 (3): >50"
```

The units of interest are “MG/L”, “Sq Ft”, “None”, and “Count”. “None” is there for the flowcode category.

0.2.7 Data Cleaning

After reviewing the original dataset and what is important for analysis, the following is the cleaning done to achieve the desired new dataset to work with.

```
# cleaned data, all managed in one spot and edited throughout the analysis
water_cleaned <- water |>
  # removing redundancy
  pivot_wider(
    # the values of the variable `parameter` will become variables
    names_from = PARAMETER,
    # the cell values values of `result` will match the corresponding variable
    values_from = RESULT) |>
  # selecting variables for analysis and changing names
  select(
    sampledate = `SAMPLE_DATE`,
    site = `SITE_NAME`,
    DO = `DISSOLVED OXYGEN`,
    Flowcode = "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)",
```

```

salamanders = "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED",
area = "AREA SAMPLED",
units = UNIT,
filter = FILTER
) |>
# fixing sampleddate format
mutate(
  sampleddate = as_datetime(mdy_hms(sampledate))
) |>
# choosing specific sites
filter(site == c("Barton Spring",
                 "Eliza Spring",
                 "Old Mill (Sunken Gardens) Spring",
                 "Upper Barton Spring"),
       # removing unused units
       units != "Minutes",
       units != "gal/min",
       units != "Cubic Feet Per Second",
       units != "1-2: (Yes; No)",
       units != "(%) Percent",
       units != "Inches",
       units != "Feet",
       units != "MM",
       units != "Feet per second")

```

0.3 Results

```
summary(water_cleaned$D0)
```

0.3.0.1 Summary Stats

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##  0.000   5.002    5.920   5.921   6.658   21.300   10087
```

```

water_cleaned |>
  group_by(filter) |>
  count(n()) |>
  select(filter, 'count' = n)

```

```

## # A tibble: 3 x 2
## # Groups:   filter [3]
##   filter    count
##   <chr>    <int>
## 1 Dissolved    454
## 2 N/A         92
## 3 Total     10019

```

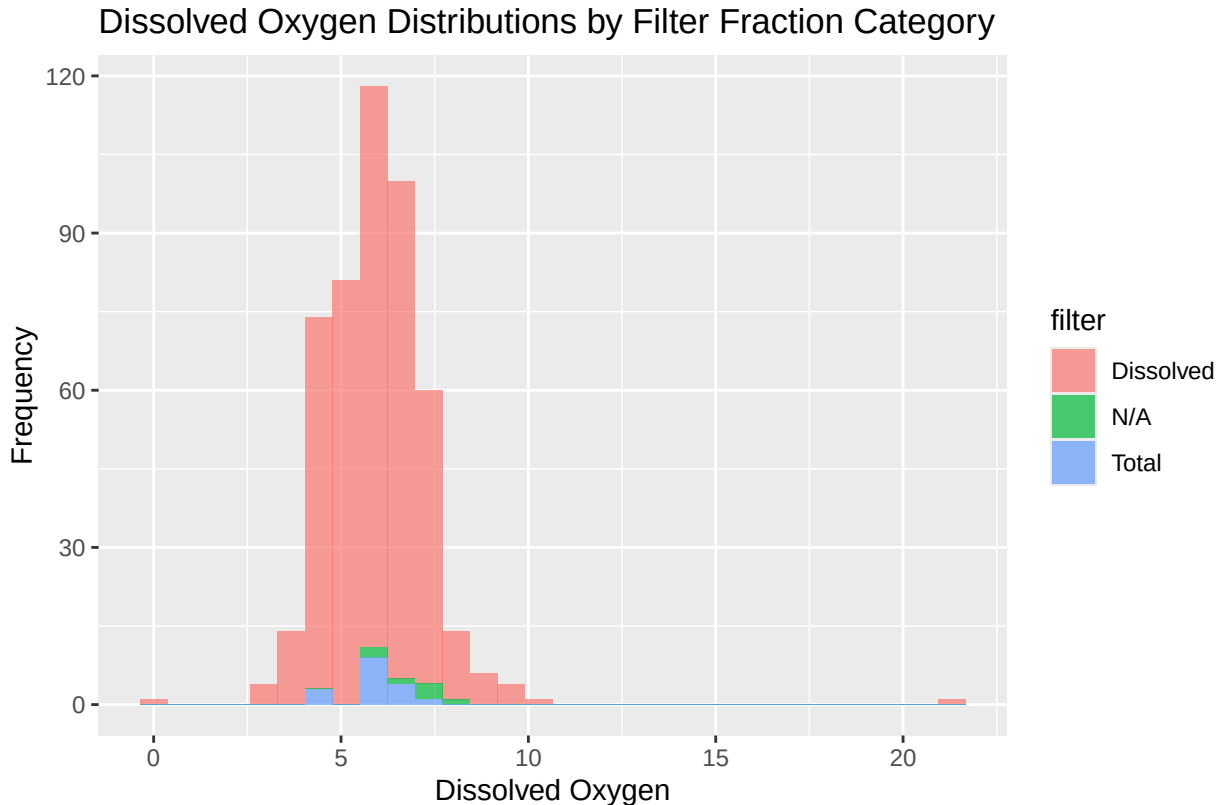
0.3.0.2 Dissolved Oxygen Given the three categories of `filter`, what are their D0 value distributions measured in mg/L?

```

water_cleaned |>
  filter(units == "MG/L") |>
  ggplot() +

```

```
geom_histogram(aes(x = DO, fill = filter), alpha = 0.7) +
labs(title = "Dissolved Oxygen Distributions by Filter Fraction Category",
      x = "Dissolved Oxygen",
      y = "Frequency",
      caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```



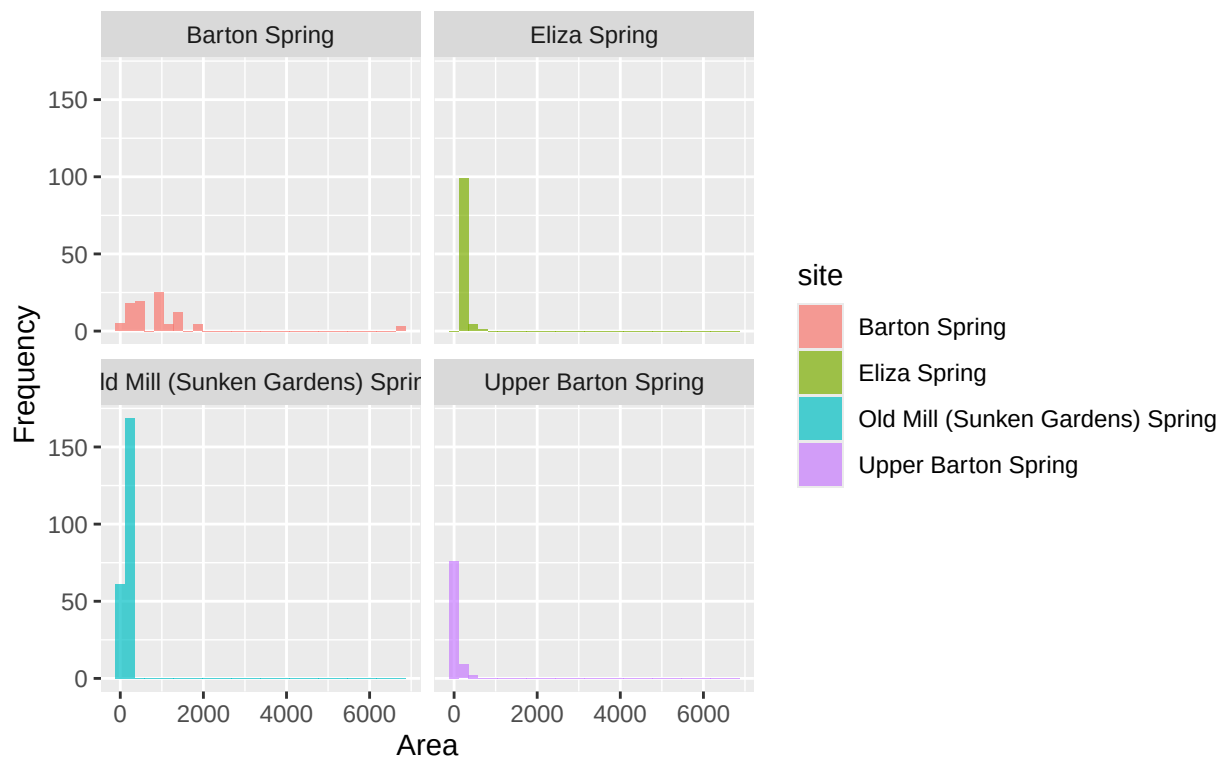
Data from the City of Austin Barton Springs Salamanders DO and Flow

This clearly shows that all of the filter fraction categories have similar ranges—excluding outliers from “Dissolved”—just that N/A and Total have fewer measurements in their pools. This makes sense because (waiting for response from MR. Clayton)

Before directly addressing how dissolved oxygen (DO) levels vary across different spring sites, and what is their relationship with seasonal changes is, it is important to know some things about the sites themselves. First, we looked at what the variation in areas sampled for each site looked like

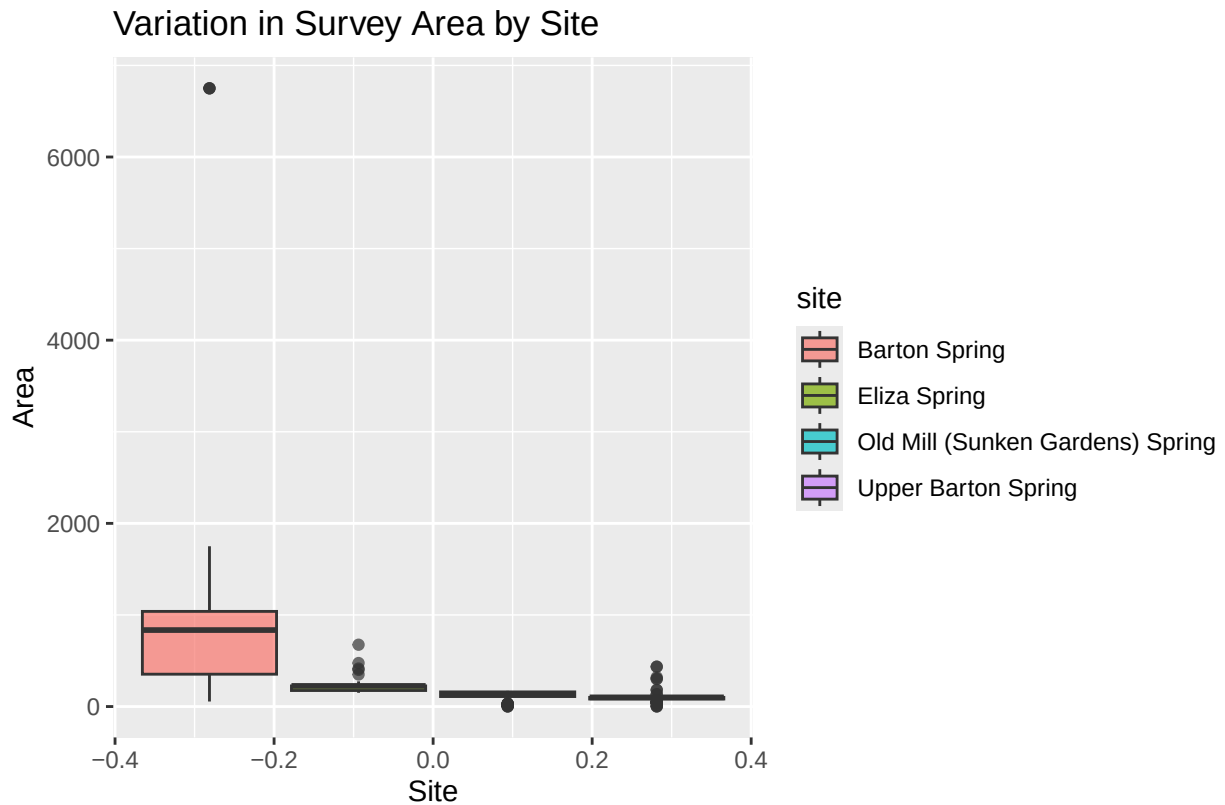
```
water_cleaned |>
  filter(units == "Sq Ft") |>
  ggplot() +
  geom_histogram(aes(x = area, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Variation in Survey Area by Site",
        x = "Area",
        y = "Frequency",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

Variation in Survey Area by Site



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
water_cleaned |>
  filter(units == "Sq Ft") |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = area, fill = site), alpha = 0.7) +
  labs(title = "Variation in Survey Area by Site",
        x = "Site",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

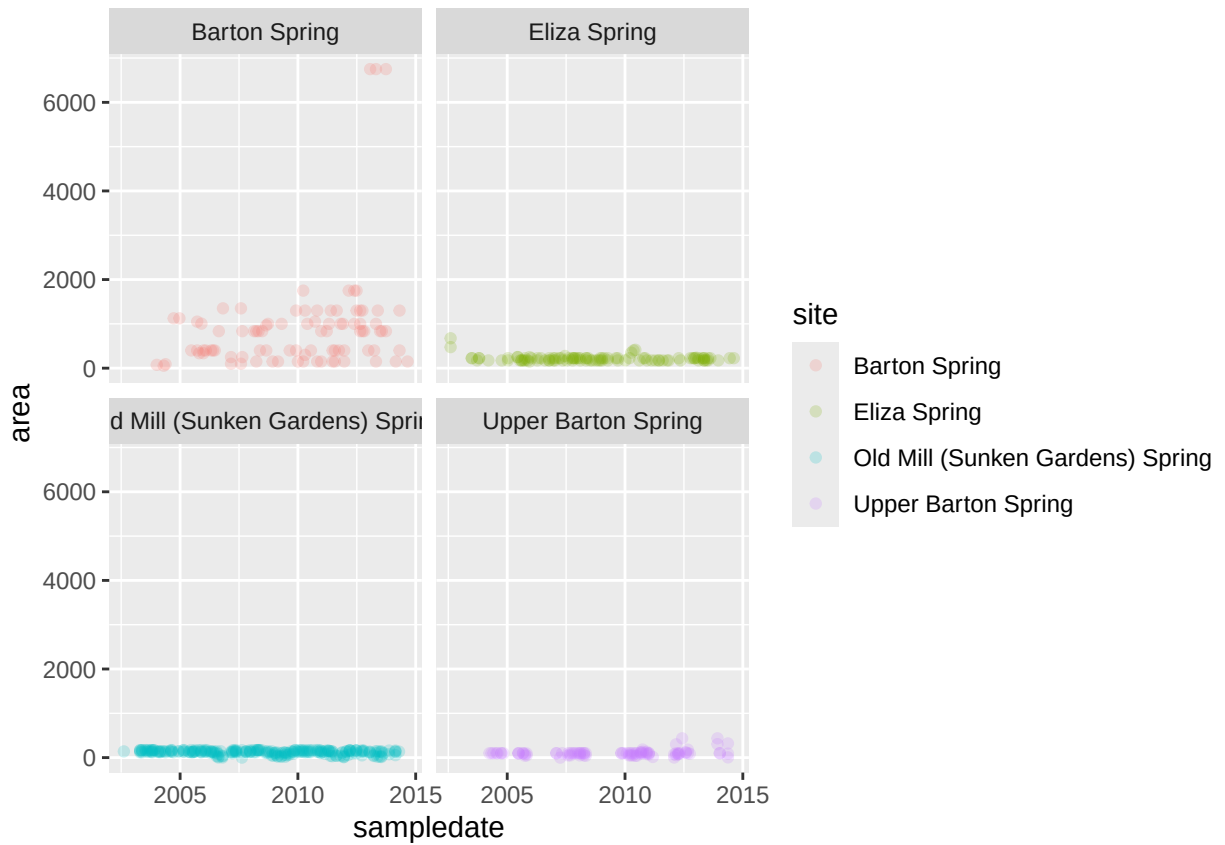



Data from the City of Austin Barton Springs Salamanders DO and Flow

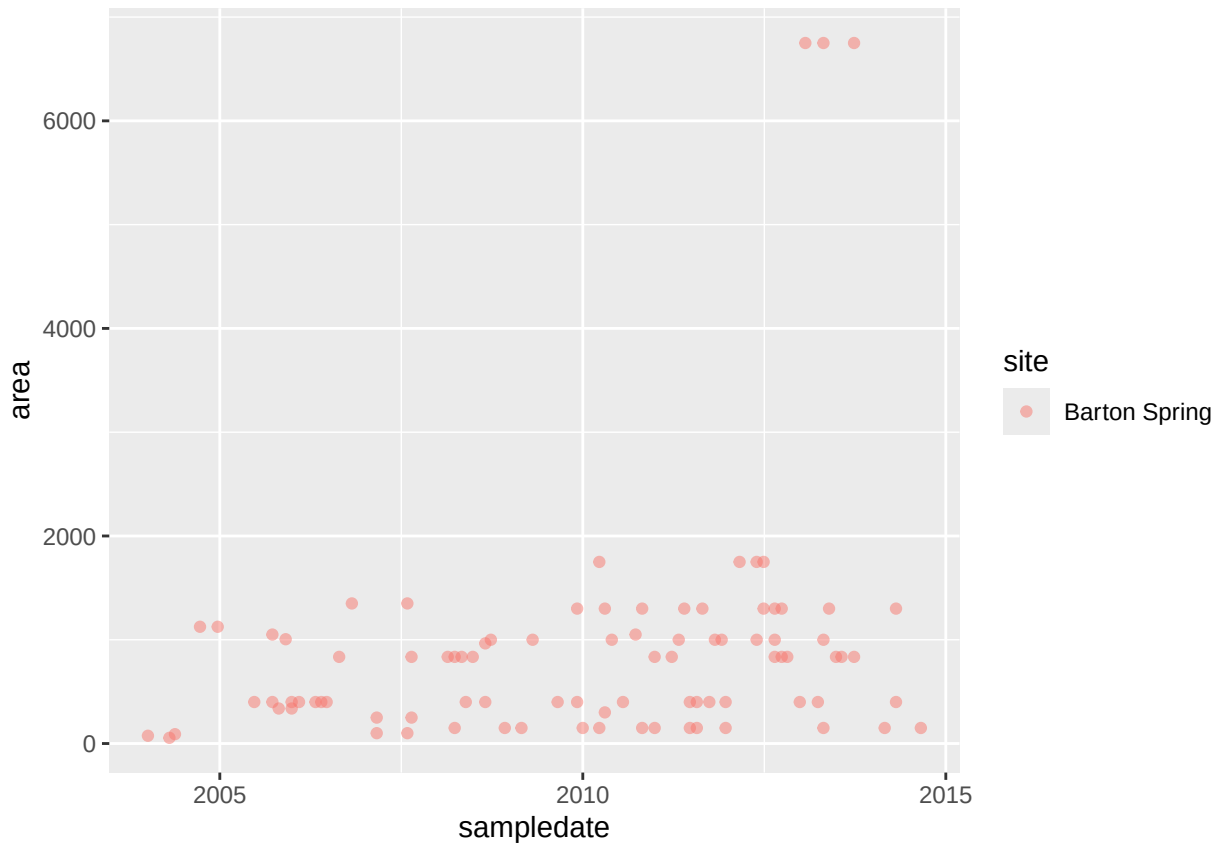
While the areas sampled for Eliza, Old Mill (Sunken Garden), and Upper Barton Springs were relatively consistent, Barton Spring's sample areas were much more spread, and also had the greatest outlier among them all.

We were then interested in the when these surveys took place, and if anything could be noticed to that helps explain the Barton Springs outliers

```
water_cleaned |>
  filter(units == "Sq Ft") |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = area, color = site), alpha = 0.2) +
  facet_wrap(~site)
```



```
water_cleaned |>
  filter(units == "Sq Ft") |>
  filter(site == "Barton Spring") |>
  ggplot() +
  geom_point(aes(x = sampledate, y = area, color = site), alpha = 0.5)
```



Taking a closer look at the 3 highest area surveys for Barton Spring

```
# Check high values of area
water_cleaned |>
  slice_max(n = 3, area)
```

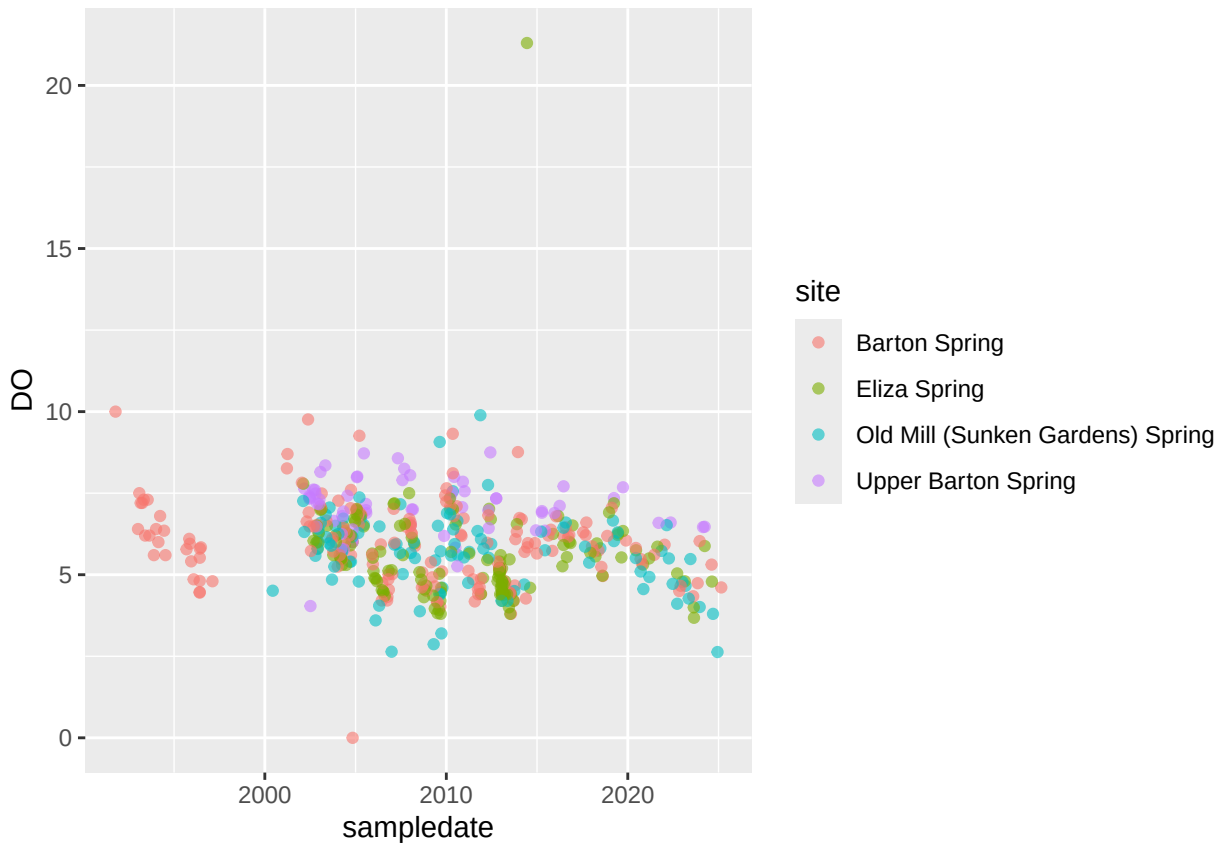
```
## # A tibble: 3 x 8
##   sampledate      site      DO Flowcode salamanders  area units filter
##   <dtm>          <chr>    <dbl>   <dbl>      <dbl> <dbl> <chr> <chr>
## 1 2013-09-26 10:50:00 Barton Spri~    NA     NA        NA  6750 Sq Ft Total
## 2 2013-04-25 11:00:00 Barton Spri~    NA     NA        NA  6750 Sq Ft Total
## 3 2013-01-24 10:50:00 Barton Spri~    NA     NA        NA  6750 Sq Ft Total
```

The highest survey area for Barton Spring is 6750 Sq Ft, taking place on the 3 different occasions listed.

With the exception of Barton Spring, most the sites have consistent survey areas. How do their dissolved oxygen levels vary over time?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampledate, y = DO, color = site), alpha = 0.6)
```

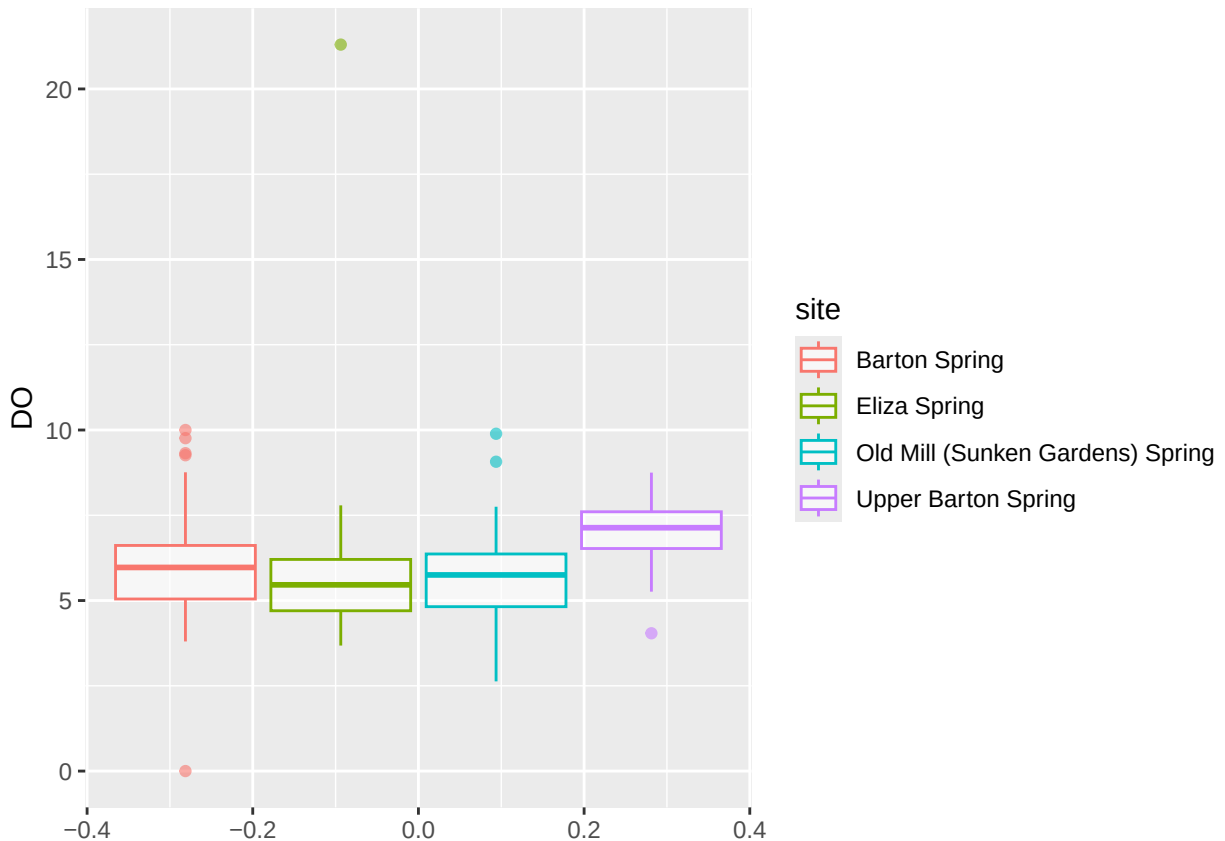
```
## Warning: Removed 10087 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



DO levels relatively similar and do not vary much over time, with a few exceptions for Barton and Eliza Springs. DO variation by site?

```
# log10(DO) by site
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = DO, color = site), alpha = 0.6)
```

```
## Warning: Removed 10087 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



DO variation for each site seems smallest for Upper Barton Spring, and a bit wider for the others.

```
water_cleaned |>
  summarize("Total salamander count" = sum(salamanders, na.rm = T))
```

0.3.0.3 Salamanders

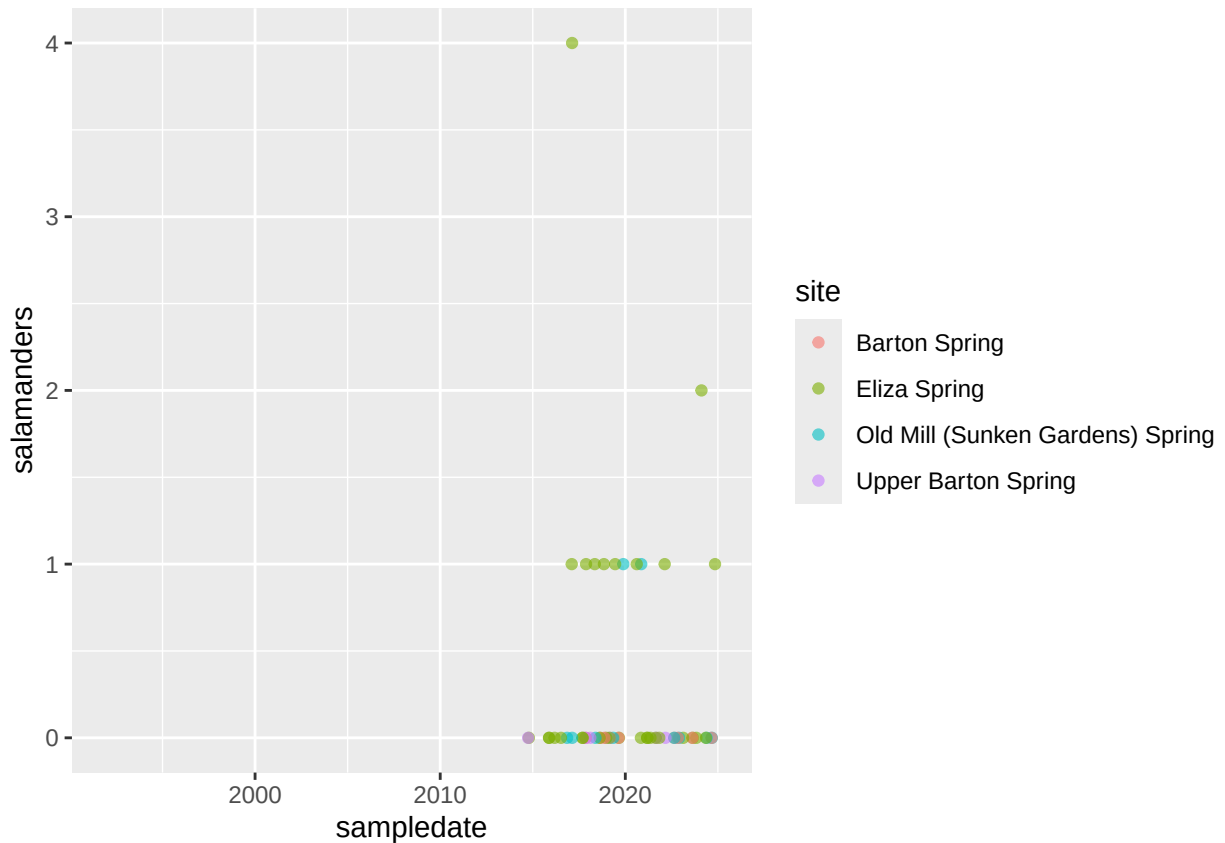
```
## # A tibble: 1 x 1
##   `Total salamander count`
##             <dbl>
## 1                16
```

A total of 16 salamanders were counted across the entire dataset. This is not to say that there exist 16 salamanders in total though, as several are likely repeated counts over time.

How is this distributed by region?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = salamanders, color = site), alpha = 0.6)
```

```
## Warning: Removed 10510 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



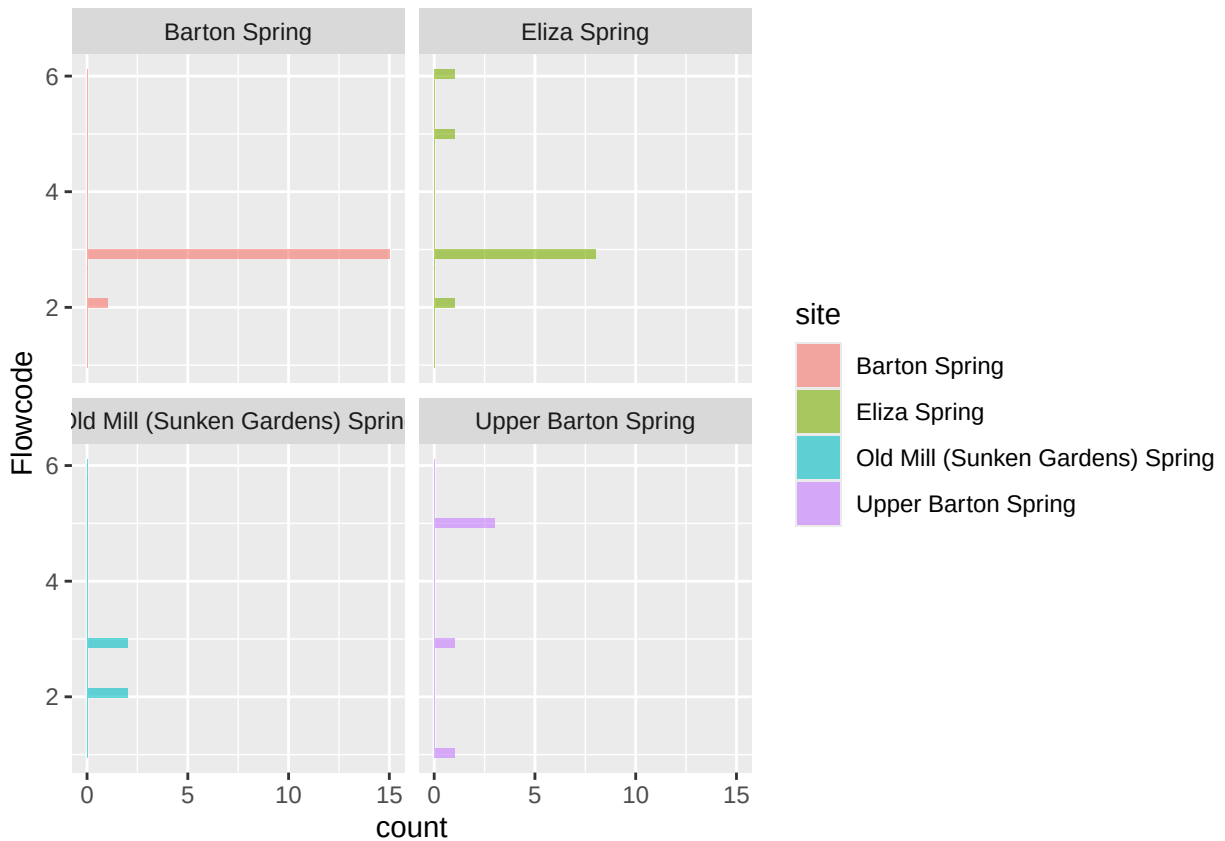
seems like the salamander counting for this view does not start until around 2015, or that there were no salamanders to count until then, and that they like Eliza Spring to most.

Barton Spring surprisingly lower than average DO when compared to the other springs.

Given this is the case, Does flow perhaps effect DO?

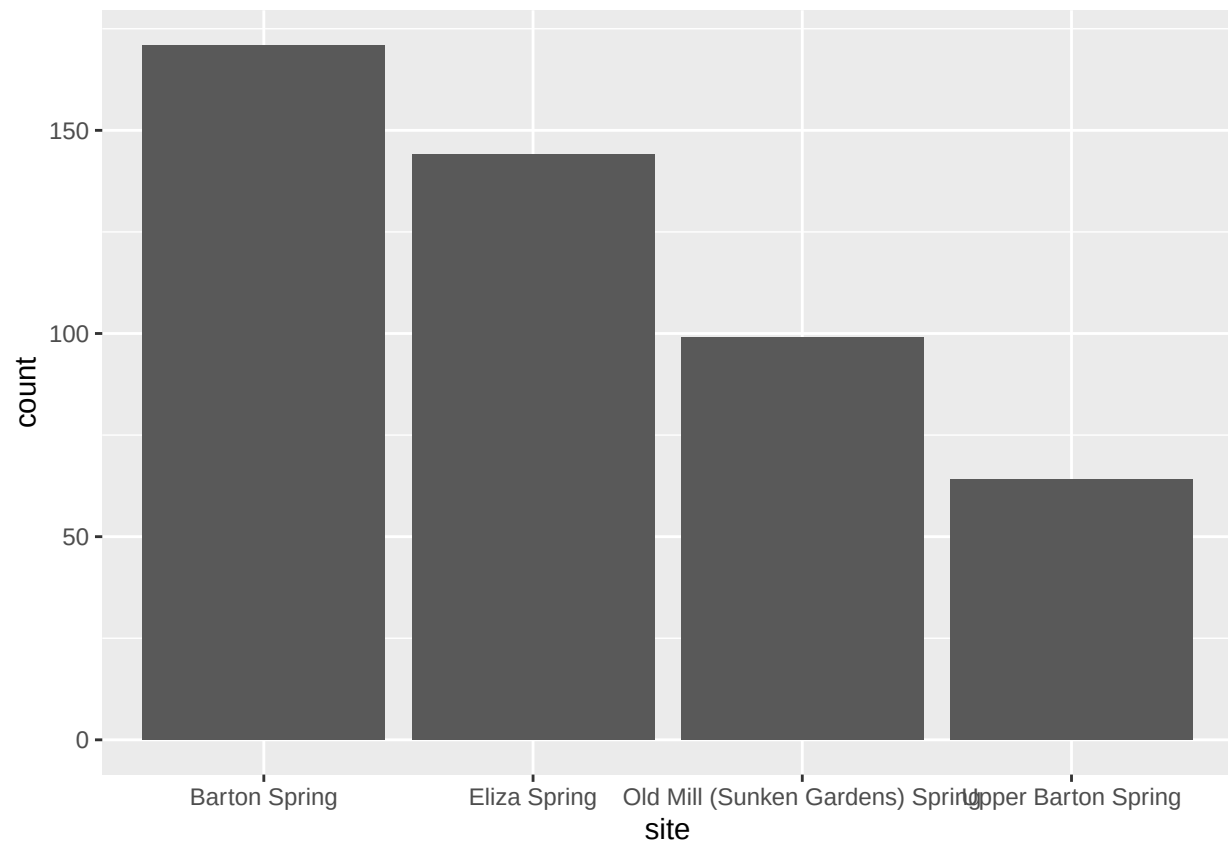
```
water_cleaned |>
  ggplot() +
  geom_histogram(aes(y = Flowcode, fill = site), alpha = 0.6) +
  facet_wrap(~site)
```

```
## Warning: Removed 10529 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



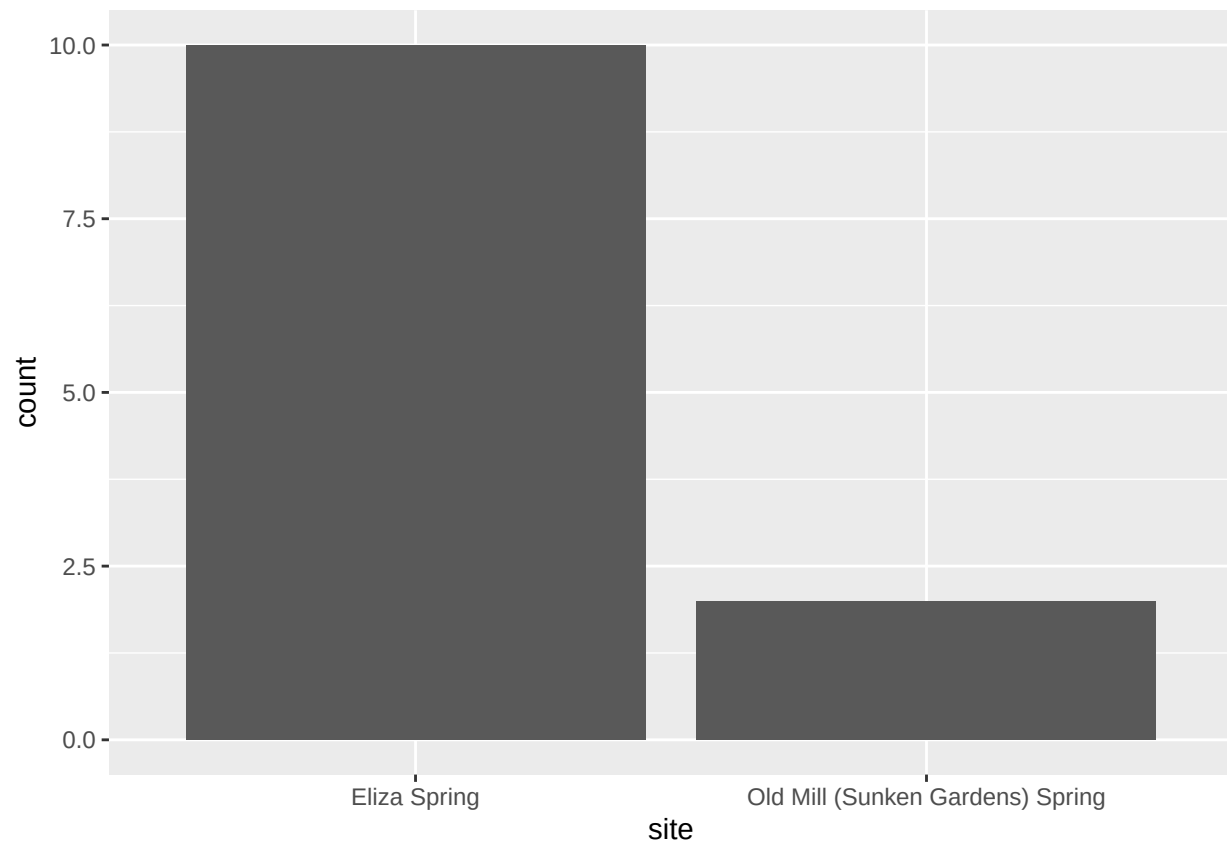
Seems a little like, for the variables we are interested in, Barton and Eliza Springs are over represented in the data. Let's check for sampling country for each

```
water_cleaned |>
  group_by(site) |>
  filter(units == "MG/L") |>
  ggplot() +
  geom_bar(aes(x = site))
```



And wouldn't you believe it, they are indeed the most sampled sites from this view. lets check where salamanders are being looked for the most

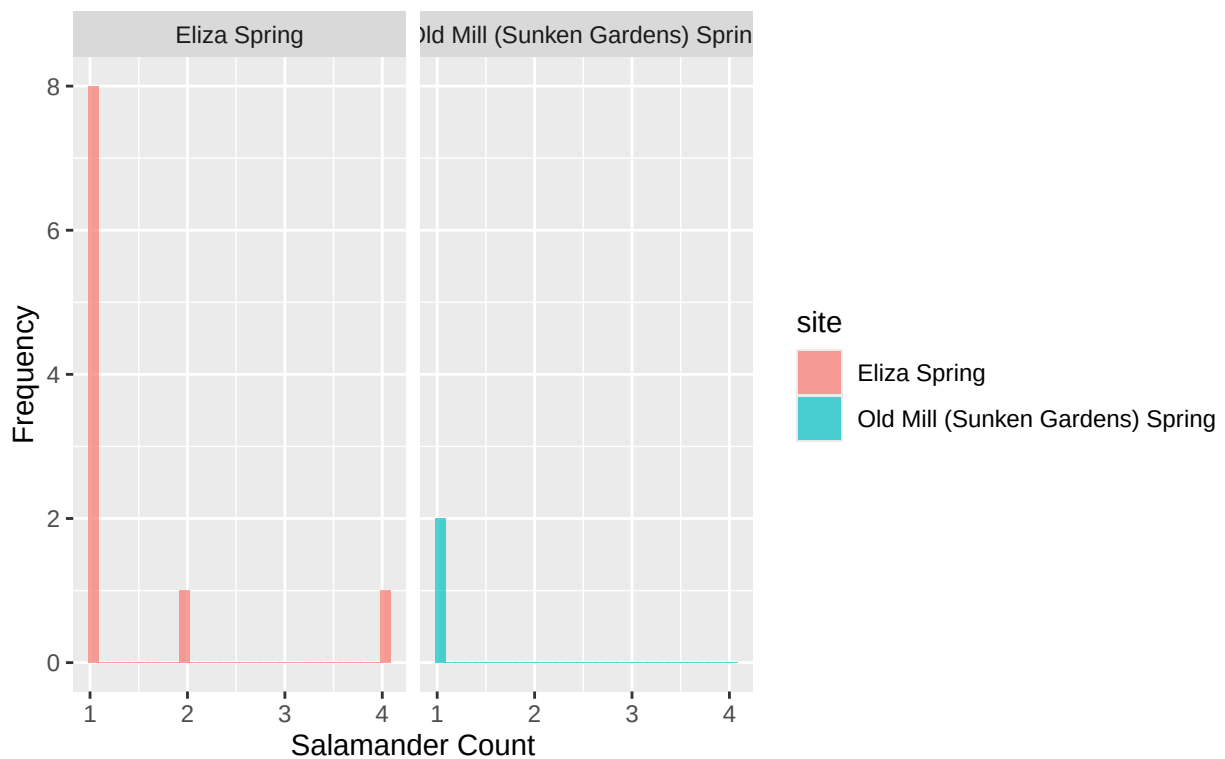
```
water_cleaned |>
  group_by(site) |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_bar(aes(x = site))
```

Well this is odd... salamander count by spring?

```
water_cleaned |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_histogram(aes(x = salamanders, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Salamander Count Distributions by Spring",
       x = "Salamander Count",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

Salamander Count Distributions by Spring

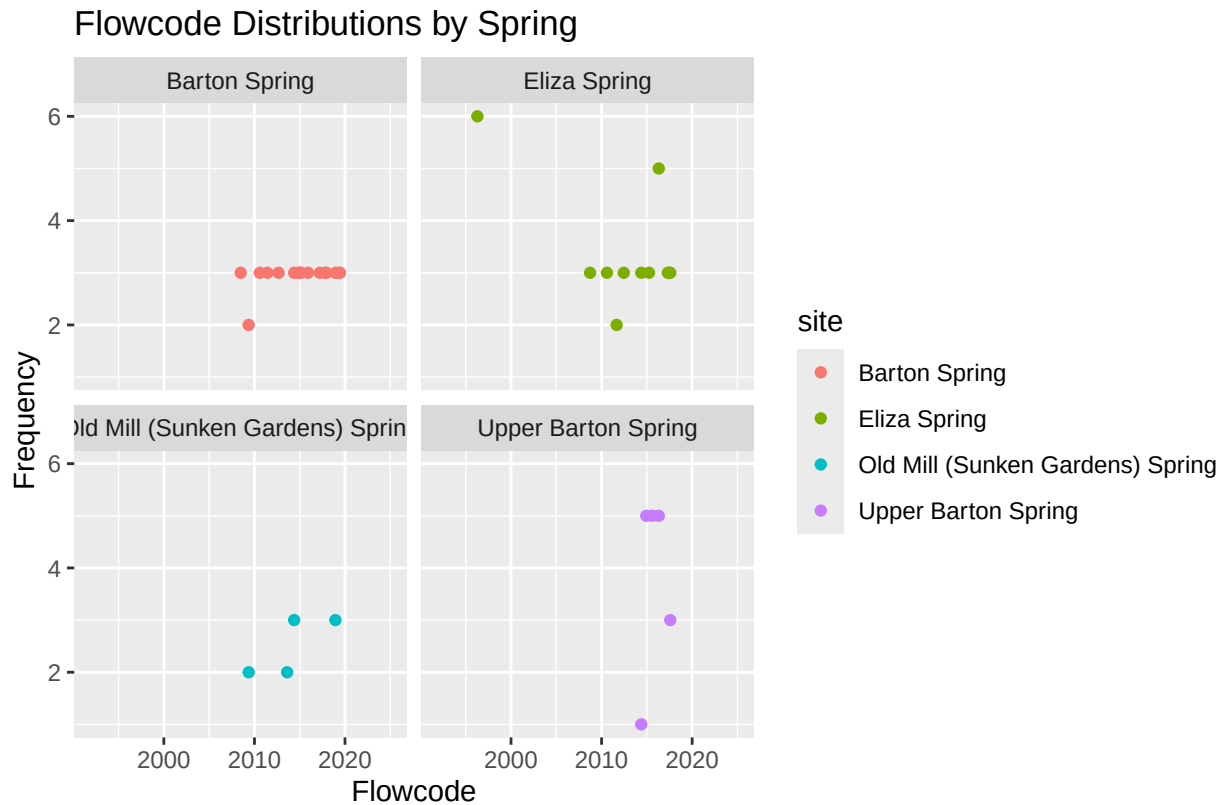


Data from the City of Austin Barton Springs Salamanders DO and Flow

Wow, there really are only 2 sites that have salamanders confirmed to be in them from this data. It's possible the other sites we chose to ignore here have more but... this is pretty interesting. I also wanted to look at flowcodes for each over time

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = Flowcode, color = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Distributions by Spring",
       x = "Flowcode",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 10529 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



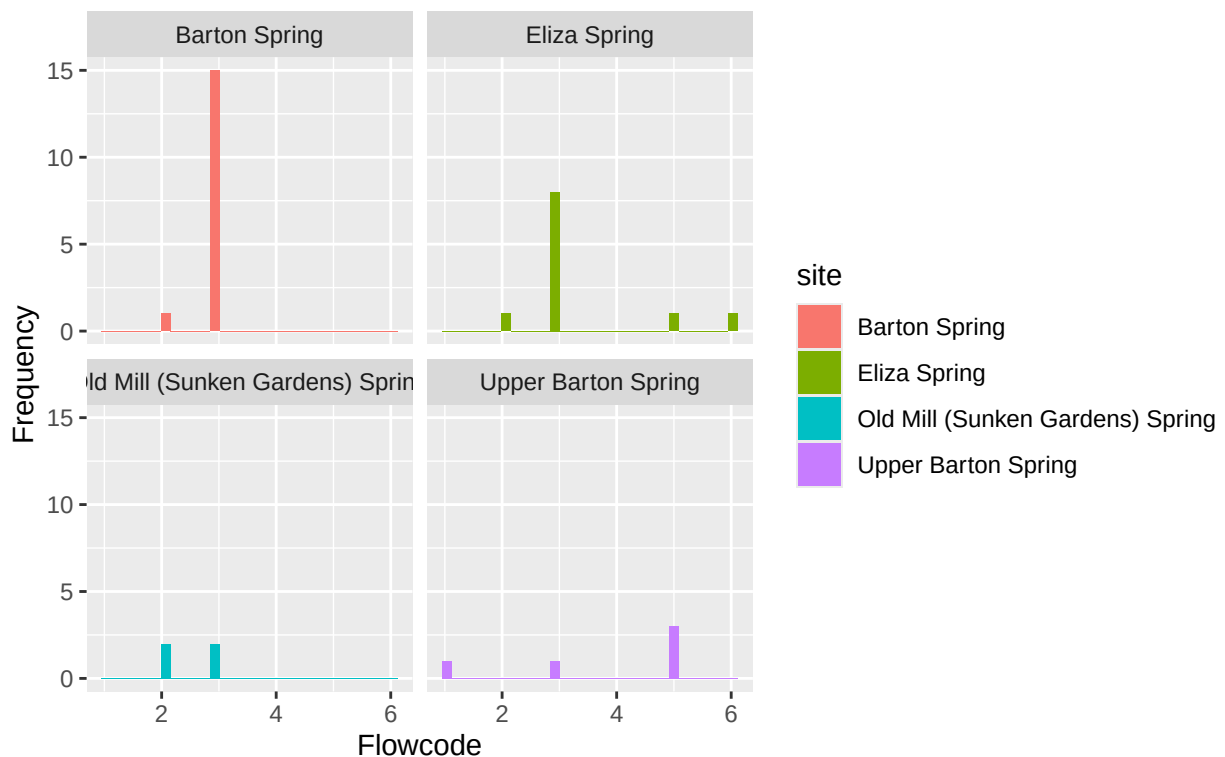
Data from the City of Austin Barton Springs Salamanders DO and Flow

typical flowcode distributions for each

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_histogram(aes(x = Flowcode, fill = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Distributions by Spring",
        x = "Flowcode",
        y = "Frequency",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 10529 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

Flowcode Distributions by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

Nick's Model

```
# Getting rid of NA values
model_water <- water_cleaned |>
  filter(!is.na(DO))
# Fit model
lin_model <- lm(DO ~ sampleddate + site, data = model_water)
summary(lin_model)
```

```
##
## Call:
## lm(formula = DO ~ sampleddate + site, data = model_water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.0440 -0.7669 -0.0166  0.6549 15.8401
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.261e+00  3.484e-01  20.841 < 2e-16 ***
## sampleddate  -1.107e-09  2.773e-10  -3.992 7.60e-05 ***
## siteEliza Spring  -2.484e-01  1.488e-01  -1.670  0.0956 .
## siteOld Mill (Sunken Gardens) Spring -1.751e-01  1.656e-01  -1.058  0.2908
## siteUpper Barton Spring  1.178e+00  1.901e-01   6.198 1.24e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.297 on 473 degrees of freedom
```

```

## Multiple R-squared:  0.1466, Adjusted R-squared:  0.1394
## F-statistic: 20.32 on 4 and 473 DF,  p-value: 1.847e-15

# Calculating RMSE
sqrt(mean(resid(lin_model)^2))

## [1] 1.290002

# Number of folds
k = 5

# Setting a seed
set.seed(322)

# Randomly order rows in the dataset
shuffled_data <- model_water[sample(nrow(model_water)),]

# Create k folds in the dataset
shuffled_data <- shuffled_data |>
  mutate(folds = cut(seq(1:nrow(shuffled_data)), breaks = k, labels = FALSE))

# Initialize a vector to keep track of the performance for each k-fold
perf_k <- NULL

# Use a for-loop to get performance for each k-fold
for(i in 1:k){
  # Split data into train and test data
  train_i <- shuffled_data |> filter(folds != i) #train data = all observations except in fold i
  test_i <- shuffled_data |> filter(folds == i) # test data = observations in fold i

  # Train model on train data (all but fold i)
  train_model <- lm(DO ~ site + sampleddate,
                    data = train_i)

  # Performance listed for each test data = fold i
  sal_per <- ggplot(test_i) +
    geom_roc(aes(
      # Outcome is DO
      d = DO,
      # Probability of Pokemon legendary status on logistic model
      m = predict(train_model, newdata = test_i)[ ,2]))
  # Save the AUC values
  perf_k[i] <- sqrt(mean(resid(train_model)^2))
}

# Performance for each fold
perf_k

## [1] 1.364279 1.343774 1.309147 1.068932 1.333599

# Average performance over all k folds
mean(perf_k)

## [1] 1.283946

My Model

```

```

# Getting rid of NA values
model_water2 <- water_cleaned |>
  filter(!is.na(salamanders))
# Fit model
lin_model2 <- lm(salamanders ~ sampleddate + site, data = model_water2)
summary(lin_model2)

##
## Call:
## lm(formula = salamanders ~ sampleddate + site, data = model_water2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.4797 -0.4435 -0.1794  0.0166  3.5333
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.857e-01  1.761e+00   0.162   0.872
## sampleddate     -1.772e-10  1.082e-09  -0.164   0.871
## siteEliza Spring   4.445e-01  2.740e-01   1.622   0.111
## siteOld Mill (Sunken Gardens) Spring 1.978e-01  3.238e-01   0.611   0.544
## siteUpper Barton Spring -1.201e-02  3.756e-01  -0.032   0.975
##
## Residual standard error: 0.6821 on 50 degrees of freedom
## Multiple R-squared:  0.08209,    Adjusted R-squared:  0.008653
## F-statistic: 1.118 on 4 and 50 DF,  p-value: 0.3586

# Calculating RMSE
sqrt(mean(resid(lin_model2)^2))

## [1] 0.6503837

# Number of folds
k = 5

# Setting a seed
set.seed(322)

# Randomly order rows in the dataset
shuffled_data <- model_water2[sample(nrow(model_water2)),]

# Create k folds in the dataset
shuffled_data <- shuffled_data |>
  mutate(folds = cut(seq(1:nrow(shuffled_data)), breaks = k, labels = FALSE))

# Initialize a vector to keep track of the performance for each k-fold
perf_k <- NULL

# Use a for-loop to get performance for each k-fold
for(i in 1:k){
  # Split data into train and test data
  train_i <- shuffled_data |> filter(folds != i) #train data = all observations except in fold i
  test_i <- shuffled_data |> filter(folds == i) # test data = observations in fold i

  # Train model on train data (all but fold i)

```

```

train_model <- lm(salamanders ~ site + sampleddate,
                  data = train_i)

# Performance listed for each test data = fold i
sal_per <- ggplot(test_i) +
  geom_roc(aes(
    # Outcome is salamanders
    d = salamanders,
    # Probability of Pokemon legendary status on logistic model
    m = predict(train_model, newdata = test_i)[,2]))
# Save the AUC values
perf_k[i] <- sqrt(mean(resid(train_model)^2))
}

# Performance for each fold
perf_k

## [1] 0.6939060 0.4300960 0.6418980 0.7137590 0.7001371

# Average performance over all k folds
mean(perf_k)

## [1] 0.6359592

water_cleaned |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_point(aes(y = salamanders, x = sampleddate, color = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Salamander Measurements by Spring",
       x = "Time",
       y = "Salamander Count",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")

```

Salamander Measurements by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
summary(water_cleaned$salamanders)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  0.000  0.000  0.000  0.291  0.000  4.000  10510
```

```
water_cleaned |>
  filter(!is.na(salamanders)) |>
  count(n())
```

```
## # A tibble: 1 x 2
##   `n()`      n
##   <int> <int>
## 1     55    55
```

0.4 Discussion

0.5 Reflection, Acknowledgements, & References. Measurement

Huge thank you to Mr.Robert Clayton, who's help was integral to this work